NONDIMENSIONALIZATION AND SIMILARITY

Fluid, transport, and immersed-structure parameters

$$\mathrm{Re} = \frac{\rho U L}{\mu}, \qquad \mathrm{St}_L = \frac{f L}{U}$$

$$\alpha = L\sqrt{\frac{\omega}{\nu}}, \qquad \mathrm{Pe} = \frac{U L}{D}$$

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IBFE Tutorials • Module 0

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

WHY NONDIMENSIONALIZE?

Nondimensional variables let us:

- compare systems with different sizes, speeds, and materials;
- identify which physical effects dominate;
- reduce the number of independent parameters;
- design dynamically similar experiments and simulations.

> A nondimensional solution can always be converted back to physical units once the reference scales are known.

Choose and report the scales. A value such as $\mathrm{Re} = 100$ is only reproducible when L and U are defined.

Background: S. Vogel, *Life in Moving Fluids* (1994) and D. J. Acheson, *Elementary Fluid Dynamics* (1990). Definitions of the Strouhal, Womersley, and Péclet numbers were adapted from Wikipedia (CC BY-SA 4.0) in the original notes. Full references are at the end of the deck.

CHARACTERISTIC SCALES

Choose scales from the physics of the problem:

Quantity	Scale	Possible choice
Length	L	chord, diameter, body length
Velocity	U	mean inflow, swim speed, tip speed
Time	$T = L/U$	advective time
Pressure	$P_I = \rho U^2$	inertial pressure

> Use the same L and U in every dimensionless group and coefficient unless a different scale is explicitly introduced.

The computational domain must be large enough that moving the outer boundary farther away does not change the quantity of interest. There is no universal "$10L$" rule.

DIMENSIONLESS VARIABLES AND DERIVATIVES

Define starred variables by

$$\boldsymbol{x} = L\boldsymbol{x}^*, \qquad \boldsymbol{u} = U\boldsymbol{u}^*, \qquad t = \frac{L}{U}t^*, \qquad p - p_0 = \rho U^2 p_I^*.$$

The differential operators transform as

$$\nabla = \frac{1}{L}\nabla^*, \qquad \frac{\partial}{\partial t} = \frac{U}{L}\frac{\partial}{\partial t^*}, \qquad \frac{D\boldsymbol{u}}{Dt} = \frac{U^2}{L}\left[\frac{\partial \boldsymbol{u}^*}{\partial t^*} + (\boldsymbol{u}^*\cdot\nabla^*)\boldsymbol{u}^*\right].$$

> A prime can be mistaken for a derivative. Stars make the dimensionless variables visually distinct throughout the derivation.

INERTIAL FORM OF INCOMPRESSIBLE NAVIER–STOKES

For a Newtonian fluid with constant density ρ and viscosity μ,

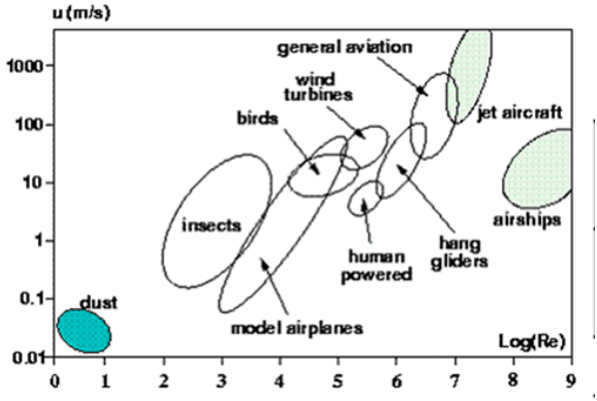
$$\boxed{\frac{\partial \boldsymbol{u}^*}{\partial t^*} + (\boldsymbol{u}^* \cdot \nabla^*)\boldsymbol{u}^* = -\nabla^* p_I^* + \frac{1}{\mathrm{Re}} \nabla^{*2} \boldsymbol{u}^* + \boldsymbol{b}^*}$$

$$\boxed{\nabla^* \cdot \boldsymbol{u}^* = 0}, \qquad \mathrm{Re} = \frac{\rho U L}{\mu} = \frac{U L}{\nu}, \qquad \nu = \frac{\mu}{\rho}.$$

Derivation: divide the dimensional momentum equation by $\rho U^2/L$; the viscous term becomes $\mu U/L^2 \div \rho U^2/L = 1/\mathrm{Re}$.

Here $\boldsymbol{b}^* = (L/U^2)\boldsymbol{b}$ is a dimensionless body-force acceleration. If $\boldsymbol{b} = -g\hat{\boldsymbol{\jmath}}$, then $\boldsymbol{b}^* = -(1/\mathrm{Fr}^2)\hat{\boldsymbol{\jmath}}$, where $\mathrm{Fr} = U/\sqrt{gL}$ is the Froude number.

REYNOLDS NUMBER



Flight speed versus $\log_{10} \mathrm{Re}$, from dust to airships. Redrawn after P. B. S. Lissaman, "Low-Reynolds-number airfoils," *Annu. Rev. Fluid Mech.* 15 (1983), 223–239, Fig. 1.

$$\mathrm{Re} = \frac{\text{inertial scale}}{\text{viscous scale}} = \frac{\rho U^2/L}{\mu U/L^2}.$$

- $\mathrm{Re} \ll 1$: viscous stresses dominate; inertia is a perturbation.
- $\mathrm{Re} \gg 1$: inertia is important; thin shear layers and separation may occur.

Each unit of the abscissa spans a decade in Re.

> Reynolds number identifies a balance. It does not, by itself, determine whether a flow is turbulent.

VISCOUS NORMALIZATION AT LOW REYNOLDS NUMBER

The inertial pressure scale is convenient when inertia is important:

$$\frac{D\boldsymbol{u}^*}{Dt^*} = -\nabla^* p_I^* + \frac{1}{\mathrm{Re}}\nabla^{*2}\boldsymbol{u}^* + \boldsymbol{b}^*.$$

For creeping-flow problems, choose the viscous pressure scale

$$p - p_0 = \frac{\mu U}{L} p_V^*.$$

The same dimensional equation then becomes

$$\boxed{\mathrm{Re}\frac{D\boldsymbol{u}^*}{Dt^*} = -\nabla^* p_V^* + \nabla^{*2}\boldsymbol{u}^* + \mathrm{Re}\,\boldsymbol{b}^*}.$$

> Both forms are valid at every Reynolds number. They satisfy $p_V^* = \mathrm{Re}\, p_I^*$, and the inertially scaled body acceleration acquires the same factor; there is no mathematical switch at $\mathrm{Re} = 1$.

DIMENSIONLESS TIME AND FREQUENCY

The advective time scale L/U gives

$$t^* = \frac{tU}{L}, \qquad f^* = \frac{fL}{U}, \qquad \omega^* = \frac{\omega L}{U} = 2\pi f^*.$$

- One dimensionless cycle has period $1/f^*$.
- Matching f^* preserves the ratio of oscillation time to advection time.
- Pumping, flapping, and pulsatile-flow comparisons must use the same definitions of L and U.

> A dimensional frequency cannot be compared across geometrically or dynamically different systems until it is combined with the relevant time scale.

FREQUENCY SCALING AND THE STROUHAL NUMBER

For vortex shedding or another oscillation with frequency f,

$$\boxed{\mathrm{St}_L = \frac{fL}{U}}.$$

This compares the oscillation time scale $1/f$ with the advective time scale L/U.
For an oscillating wake, wing, or propulsor,

$$\boxed{\mathrm{St}_A = \frac{fA}{U}},$$

where A is the stated excursion scale. If $A = L$, then $\mathrm{St}_A = \mathrm{St}_L$.

> State whether the characteristic length is a chord, diameter, wake width, or motion amplitude. Different choices define different Strouhal numbers.

The number is named for V. Strouhal, who studied vortex shedding from wires in 1878. Definition adapted from Wikipedia, “Strouhal number” (CC BY-SA 4.0), as in the original notes.

STROUHAL NUMBER IN SWIMMING AND FLYING

For a swimming or flying animal, a common kinematic definition is

$$\boxed{\mathrm{St}_A = \frac{fA}{U}},$$

where f is stroke or tail-beat frequency, A is a stated excursion amplitude, and U is mean forward speed.

- St_A organizes vortex-growth and shedding regimes for pitching and heaving propulsors.
- Many cruising swimmers and flyers in the Taylor et al. data lie in $0.2 < \mathrm{St}_A < 0.4$.
- The interval is empirical, not a universal efficiency optimum for every body, gait, or amplitude convention.

State whether A is peak or peak-to-peak excursion; Taylor et al. use the peak-to-peak excursion of the wingtip or tail tip.

Taylor, Nudds, and Thomas, *Nature* 425 (2003), 707–711.

WOMERSLEY NUMBER

For oscillatory viscous flow with angular frequency ω,

$$\boxed{\alpha = L\sqrt{\frac{\omega}{\nu}}}.$$

- $\alpha \ll 1$: viscosity communicates the oscillation across the length scale during each cycle.
- $\alpha \gg 1$: unsteady inertia dominates away from a thin oscillatory boundary layer.

$$\delta_\omega \sim \sqrt{\frac{2\nu}{\omega}}, \qquad \alpha = \sqrt{2}\,\frac{L}{\delta_\omega}.$$

If Re and $\mathrm{St}_L = fL/U$ use the same L and U,

$$\boxed{\alpha^2 = 2\pi\,\mathrm{Re}\,\mathrm{St}_L}.$$

If one group uses pipe radius and another uses diameter, the relation acquires the corresponding factor.

Interpretation: $\alpha^2 = \omega L^2/\nu$ is the ratio of transient inertia, $\rho\omega U$, to viscous force, $\mu U/L^2$. Named for J. R. Womersley (*J. Physiol.* 127, 1955). Definition adapted from Wikipedia, “Womersley number” (CC BY-SA 4.0), as in the original notes.

ILLUSTRATIVE WOMERSLEY NUMBERS IN CANINE VESSELS

The original tutorial used the following values for a canine cardiovascular system at a heart rate near $2\,\mathrm{Hz}$. Here the length scale in $\alpha = L\sqrt{\omega/\nu}$ is the vessel radius, $L = R$.

Vessel	α
Ascending aorta	13.2
Descending aorta	11.5
Abdominal aorta	8.0
Femoral artery	3.5
Carotid artery	4.4

Vessel	α
Arterioles	0.04
Capillaries	0.005
Venules	0.035
Inferior vena cava	8.8
Main pulmonary artery	15

> These values illustrate the strong vessel-size dependence of α. They are not universal physiological constants: radius, frequency, and viscosity must accompany any reported value.

Values after W. W. Nichols and M. F. O'Rourke, *McDonald's Blood Flow in Arteries*, 5th ed. (Hodder Arnold, 2005). Further sources on the next slide.

CANINE WOMERSLEY NUMBERS: A WORKED CHECK AND SOURCES

Order-of-magnitude check for the ascending aorta. Take an illustrative radius $R \approx 0.75\,\mathrm{cm}$, heart rate $f = 2\,\mathrm{Hz}$, and blood kinematic viscosity $\nu \approx 4 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$:

$$\omega = 2\pi f \approx 12.6\,\mathrm{s^{-1}}, \qquad \alpha = R\sqrt{\frac{\omega}{\nu}} \approx 7.5 \times 10^{-3}\,\mathrm{m} \times \sqrt{\frac{12.6}{4 \times 10^{-6}}}\,\mathrm{m^{-1}} \approx 13,$$

consistent with the tabulated 13.2. With the diameter in place of the radius, the same vessel would give $\alpha \approx 27$; always state which length is used.

> Because $\alpha \propto R$, a factor of $\sim 10^3$ in radius between the aorta and a capillary produces the factor of $\sim 10^3$ in α seen in the table.

Sources: the same ten canine values are commonly cited from the table of typical canine vascular parameters in C. G. Caro, T. J. Pedley, R. C. Schroter, and W. A. Seed, *The Mechanics of the Circulation* (Oxford Univ. Press, 1978), as reproduced in Wikipedia, “Womersley number” (CC BY-SA 4.0); table number to be confirmed. The original tutorial’s Womersley material was adapted from that Wikipedia article.

PÉCLET NUMBER

For a scalar concentration c with molecular diffusivity D,

$$\frac{\partial c}{\partial t} + \boldsymbol{u} \cdot \nabla c = D \nabla^2 c.$$

With $c - c_0 = C c^*$ and the same L, U, and L/U scales,

$$\frac{\partial c^*}{\partial t^*} + \boldsymbol{u}^* \cdot \nabla^* c^* = \frac{1}{\mathrm{Pe}} \nabla^{*2} c^*, \qquad \boxed{\mathrm{Pe} = \frac{UL}{D}}.$$

$\mathrm{Pe} \ll 1$: diffusion dominates transport. $\mathrm{Pe} \gg 1$: advection dominates transport.

For mass transfer, $\mathrm{Pe} = \mathrm{Re}\,\mathrm{Sc}$; for heat transfer, $\mathrm{Pe} = \mathrm{Re}\,\mathrm{Pr}$, provided all groups use the same L and U.

Here $\mathrm{Sc} = \nu/D$ and $\mathrm{Pr} = \nu/\kappa$, with κ the thermal diffusivity. Named for J. C. E. Péclet. Definitions adapted from Wikipedia, "Péclet number" (CC BY-SA 4.0), as in the original notes.

FORCE SCALES FOR A 2-D MODEL

Suppose $\mathcal{F}$ denotes a force resultant per unit out-of-plane depth, with units $\mathrm{N\,m^{-1}}$. For example, it may be the integral of boundary traction over a length of order L.

Normalization	Stress scale	Resultant scale
Inertial	ρU^2	$\mathcal{F}_I = \rho U^2 L$
Viscous	$\mu U/L$	$\mathcal{F}_V = \mu U$

$$\mathcal{F}_I^* = \frac{\mathcal{F}}{\rho U^2 L}, \qquad \mathcal{F}_V^* = \frac{\mathcal{F}}{\mu U}, \qquad \boxed{\mathcal{F}_V^* = \mathrm{Re}\,\mathcal{F}_I^*}.$$

A local force per unit boundary length and per unit depth has units $\mathrm{N\,m^{-2}}$ and scales with ρU^2 or $\mu U/L$. Begin with the force law and its measure.

DIMENSIONLESS BENDING COEFFICIENT IN TWO DIMENSIONS

Under a two-dimensional per-unit-depth convention, let k_b have units

$$[k_b] = \mathrm{N\,m}.$$

The inertial and viscous normalizations are

$$\boxed{k_b^{(I)} = \frac{k_b}{\rho U^2 L^3}}, \qquad \boxed{k_b^{(V)} = \frac{k_b}{\mu U L^2}}.$$

They describe the same dimensional coefficient and satisfy

$$k_b^{(V)} = \mathrm{Re}\, k_b^{(I)}.$$

> There is no mathematical switch at $\mathrm{Re} = 1$. Choose the normalization that exposes the balance of interest, then use it consistently.

DIMENSIONLESS STRETCHING COEFFICIENT

Let k_s denote a force-like fiber coefficient per unit out-of-plane depth:

$$[k_s] = \mathrm{N\,m^{-1}}.$$

The corresponding nondimensional coefficients are

$$\boxed{k_s^{(I)} = \frac{k_s}{\rho U^2 L}}, \qquad \boxed{k_s^{(V)} = \frac{k_s}{\mu U}},$$

with

$$k_s^{(V)} = \mathrm{Re}\, k_s^{(I)}.$$

> A point-spring stiffness and three-dimensional line, surface, or volume coefficients have different units. Begin with the exact force law and integration measure.

VISCOSITY AND REYNOLDS-NUMBER MATCHING

Because

$$\mathrm{Re} = \frac{\rho U L}{\mu} = \frac{U L}{\nu},$$

one can change Reynolds number by changing U, L, ρ, or μ.

- Varying μ is common in simulations and laboratory scale models.
- Changing viscosity may also change other material or transport parameters.
- Reporting only μ does not establish dynamic similarity; report Re and the chosen L and U.

> When an oscillation or scalar field matters, matching Reynolds number alone is insufficient. Match the relevant Strouhal, Womersley, Péclet, and structural groups as well.

FLEXURAL RIGIDITY AND EFFECTIVE STIFFNESS

For a physical three-dimensional beam,

$$\boxed{B = EI},$$

where

- E is Young's modulus $[\mathrm{N\,m^{-2}}]$;
- I is a second moment of area $[\mathrm{m^4}]$;
- EI is flexural rigidity $[\mathrm{N\,m^2}]$.

> Flexural rigidity is a material-and-section property. The force/displacement stiffness also depends on beam length, loading, and boundary conditions; commonly $K_{\mathrm{eff}} \sim EI/L_b^3$.

A 2-D model may use rigidity per unit out-of-plane depth, with units $\mathrm{N\,m}$. State that convention explicitly.

APPROXIMATE YOUNG'S MODULUS: COMPLIANT MATERIALS

Material	GPa	psi
Rubber, small strain	0.01–0.1	1,450–14,504
Low-density polyethylene	0.11–0.45	16,000–65,000
Diatom frustules, largely hydrated silica	0.35–2.77	50,000–400,000
PTFE	0.5	72,500
High-density polyethylene	0.8	116,000
Bacteriophage capsids	1–3	145,000–435,000
Polypropylene	1.5–2	218,000–290,000
Polyethylene terephthalate	2.2–2.7	320,000–390,000
Nylon	2–4	290,000–580,000
Polystyrene	3–3.5	440,000–510,000
Medium-density fiberboard	4	580,000
Oak, along the grain	11	1.60×10^6
Human cortical bone	14	2.03×10^6

These are order-of-magnitude reference values. Modulus depends on composition, orientation, hydration, strain, and test protocol.

Conversion: $1\ \mathrm{GPa} \approx 145{,}038\ \mathrm{psi}$. Values follow the table "Approximate Young's modulus for various materials" in Wikipedia, "Young's modulus" (CC BY-SA 4.0), which cites a primary source for each entry; source to be confirmed.

APPROXIMATE YOUNG'S MODULUS: STIFFER MATERIALS

Material	GPa	psi
Glass-reinforced polyester matrix	17.2	2.49×10^6
Aromatic peptide nanotubes	19–27	$2.76–3.92 \times 10^6$
High-strength concrete	30	4.35×10^6
Carbon-fiber-reinforced plastic, 50/50 biaxial	30–50	$4.35–7.25 \times 10^6$
Hemp fiber	35	5.08×10^6
Magnesium	45	6.53×10^6
Glass	50–90	$7.25–13.1 \times 10^6$
Flax fiber	58	8.41×10^6
Aluminum	69	10.0×10^6
Mother-of-pearl, largely calcium carbonate	70	10.2×10^6
Aramid	70.5–112.4	$10.2–16.3 \times 10^6$
Tooth enamel, largely calcium phosphate	83	12.0×10^6

Use a source appropriate to the actual material and loading direction before converting E into flexural rigidity EI.

Conversion: $1\ \mathrm{GPa} \approx 145{,}038\ \mathrm{psi}$. Values follow the same Wikipedia "Young's modulus" table (CC BY-SA 4.0); source to be confirmed.

SECOND MOMENT OF AREA

For a cross-sectional area A and an in-plane reference axis ℓ,

$$\boxed{I_\ell = \iint_A r_\perp^2 \, dA},$$

where $r_\perp$ is the perpendicular distance from the axis to the area element dA.

For the Cartesian axes,

$$I_x = \iint_A y^2 \, dA, \qquad I_y = \iint_A x^2 \, dA.$$

The polar second moment about the perpendicular z axis is

$$J_z = \iint_A (x^2 + y^2) \, dA = I_x + I_y.$$

> I_x and I_y control bending about the corresponding axes. J_z is a polar area moment and should not be confused with a mass moment of inertia.

The axis is written ℓ to avoid confusion with the flexural rigidity $B = EI$. For an axis parallel to a centroidal axis at distance d, the parallel-axis theorem gives $I_\ell = I_{\text{centroid}} + Ad^2$.

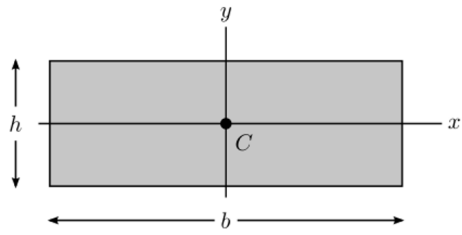
RECTANGLE WITH CENTROID AT THE ORIGIN

For width b in the x direction and height h in the y direction,

$$I_x = \int_{-b/2}^{b/2} \int_{-h/2}^{h/2} y^2 \, dy \, dx = \frac{bh^3}{12},$$

$$I_y = \int_{-h/2}^{h/2} \int_{-b/2}^{b/2} x^2 \, dx \, dy = \frac{b^3 h}{12},$$

$$J_z = I_x + I_y = \frac{bh}{12}(b^2 + h^2).$$



Rectangular cross-section of width b and height h with centroid C (schematic). Schematic: instructor's course notes.

Bending about x (load along y) uses $I_x \propto h^3$: a plank is far stiffer on edge than flat.

CONVERTING STRUCTURAL PARAMETERS

1. Select and document one set of L and U.
2. Choose inertial or viscous normalization for the full coupled model.
3. Convert the desired dimensionless coefficient back to physical units.

For example, with inertial normalization,

$$k_{\text{spring}} = k^*_{\text{spring},I}\,\rho U^2 L, \qquad k_{\text{beam}} = k^*_{\text{beam},I}\,\rho U^2 L^3.$$

In the notation of the earlier slides, $k^*_{\text{spring},I} = k_s^{(I)}$ and $k^*_{\text{beam},I} = k_b^{(I)}$ (2-D, per unit depth).

> A coefficient does not determine deformation by itself. Geometry, loading, boundary conditions, and the constitutive law all matter, so report a sensitivity study rather than a universal “good” value.

BEAM-STIFFNESS STARTING ESTIMATE

With inertial normalization,

$$k_b = k_b^{(I)} \rho U^2 L^3.$$

The original course heuristic used $k_b^{(I)} \approx 10$ as a trial value that produced modest bending, roughly a few degrees, in a particular model.

- Use the same characteristic velocity and length as the fluid scaling; a peak velocity may be appropriate for a peak-load estimate.
- In practice: try $U_{\max}$ for the velocity scale and the boundary length for L; these are rough estimates and should be reported with the choice made. If Re was defined with a different velocity U (same L), the two coefficients for one dimensional k_b satisfy $k_b^{(I)}\big|_{U_{\max}} = k_b^{(I)}\big|_{U} (U/U_{\max})^2$.
- Compare the resulting k_b with a physical rigidity such as EI under the model's per-depth convention.
- Sweep $k_b^{(I)}$ and demonstrate convergence of the deformation or performance metric.

> The numerical value 10 is a model-specific starting guess, not a universal relation between dimensionless stiffness and bending angle.

PHYSICAL LOAD ESTIMATES

At moderate or high Reynolds number, estimate a total hydrodynamic load from a physical area:

$$F_{\mathrm{char}} \sim \frac{1}{2} C_D \rho U^2 A.$$

For a 2-D model per unit depth, use the corresponding projected length instead of area. At low Reynolds number, use a geometry-specific viscous drag law.

For a target displacement $\delta_{\max}$,

$$K_{\mathrm{eff}} \sim \frac{F_{\mathrm{char}}}{\delta_{\max}}$$

provides an order-of-magnitude starting point.

> Do not use the Eulerian grid spacing Δx as a physical area or mass unless you are explicitly constructing a discrete per-node quantity. Otherwise the physical model changes when the grid is refined.

SPRING-STIFFNESS STARTING ESTIMATE

If F_b is the characteristic load supported by a spring-like constraint and $\delta_{\max}$ is the largest acceptable displacement,

$$\boxed{K_{\text{eff}} \sim \frac{F_b}{\delta_{\max}}}.$$

Choose $\delta_{\max}$ from the physical tolerance, such as maximum displacement from a target point or resting length.

At moderate or high Reynolds number,

$$F_b \sim \tfrac{1}{2} C_D \rho U^2 A.$$

At low Reynolds number, use a geometry-specific viscous drag law rather than a universal grid-based estimate.

> Convert the resulting stiffness using the units of the implemented force law. A node, line, surface, and volume spring require different coefficients.

DAMPING IN A COUPLED FLUID–STRUCTURE SYSTEM

For one linear degree of freedom,

$$m\ddot{x} + c\dot{x} + kx = 0, \qquad c_{\mathrm{crit}} = 2\sqrt{km}.$$

This formula is a useful analogy, but an immersed structure has distributed modes and frequency-dependent fluid loading.

- A massless boundary still experiences added mass and hydrodynamic damping.
- The effective mass is generally not $\rho(\Delta x)^d$.
- Select penalty damping with a modal estimate or solver-specific stability and convergence tests.

> Report the damping law and demonstrate that the physical response and time-step restriction are adequately resolved.

Critical damping, $c = c_{\mathrm{crit}}$, returns the oscillator to rest fastest without overshoot; the damping ratio is $\zeta_d = c/c_{\mathrm{crit}}$.

Background: Wikipedia, “Damping” (CC BY-SA 4.0), linked in the original notes.

EFFICIENCY AT STEADY SWIMMING OR FLYING

In a statistically steady or time-periodic state, mean kinetic energy does not grow, although it may oscillate instantaneously. Input power still balances drag and energizes the wake.
A thermodynamic efficiency (useful work, i.e. change in kinetic or potential energy, divided by energy input) would therefore be zero for every steady swimmer. Efficiency must instead be defined through a chosen performance metric, as below.

Constrained propulsor

A common definition is

$$\eta_{\mathrm{prop}} = \frac{TU}{P_{\mathrm{in}}},$$

where T is mean thrust.

Self-propelled body

Mean net force is zero. One alternative is

$$\eta_{\mathrm{QP}} = \frac{D_{\mathrm{tow}}U}{P_{\mathrm{in}}}.$$

Here D_{tow} is the drag from a matched towing test.

> State the system boundary, useful-power definition, and treatment of negative actuator work; different efficiency measures answer different questions.

For a review of thrust, drag, and efficiency definitions in swimming, see Smits, *J. Fluid Mech.* 874 (2019), P1.

COST OF TRANSPORT

If E_{in} is the input energy used to travel distance d,

$$C = \frac{E_{\mathrm{in}}}{d} = \frac{\overline{P}_{\mathrm{in}}}{\overline{U}} \quad [\mathrm{J\,m^{-1}} = \mathrm{N}].$$

Common normalizations are

$$C_m = \frac{E_{\mathrm{in}}}{md} = \frac{\overline{P}_{\mathrm{in}}}{m\overline{U}} \quad [\mathrm{J\,kg^{-1}\,m^{-1}}], \qquad C^* = \frac{\overline{P}_{\mathrm{in}}}{mg\overline{U}} \quad \text{(dimensionless, weight-normalized)}.$$

For a distributed actuator, compute input power from the work rate using a stated sign convention, for example

$$P_{\mathrm{in}}(t) = \int_{\Gamma} \boldsymbol{f}_{\mathrm{act}} \cdot \boldsymbol{u}_b \, d\Gamma,$$

where $\boldsymbol{f}_{\mathrm{act}}$ is the actuator force density on the structure Γ and $\boldsymbol{u}_b$ is the boundary velocity. Average P_{in} over a whole number of cycles before forming C.

Recall: work = force × distance [N m]; power = force × velocity [N m s^{-1}]. The mass-specific cost C_m is the form compared across animals by K. Schmidt-Nielsen, *Science* 177 (1972), 222–228; the weight-normalized C^* is the "specific resistance" of G. Gabrielli and T. von Kármán, *Mech. Eng.* 72 (1950), 775–781.

NONDIMENSIONALIZATION CHECKLIST

1. Define L, U, and every additional material or transport scale.
2. Substitute the scaled variables into the dimensional equations.
3. Choose one normalization that exposes the balance of interest.
4. Check every dimensionless coefficient by units and limiting cases.
5. Preserve the same definitions when comparing cases or redimensionalizing results.

> Dynamic similarity requires matching every dimensionless group that materially affects the problem, not only Reynolds number.

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Figure credits: `Re.png` redrawn after Lissaman (1983), Fig. 1; `IJ.png` schematic from the instructor’s course notes.